

Restricted entangler obstruction

MQM 0.5 | 18 September 2026 | Source-reported finite enumeration

Conditional finite-search lemma. For two copies of the supplied eight-data-qubit subsystem code, none of the gauge-preserving maps in the three enumerated transversal/pairwise Clifford families below induces an entangling logical map, according to the supplied search records.

Enumerated family	Tested	Preserve	Entangle
Identical two-qubit Clifford symplectic action on all eight corresponding cross-block pairs	720	2	0
Pair-specific I, CNOT A-to-B, CNOT B-to-A, CZ	65,536	1	0
Same pair-specific alphabet, plus SWAP	390,625	2	0

Code specification. Each block has center generators $IIIXZZI$, $IIYIYZZ$, $IIIXZIXZ$, $XXXXYYIX$; six further gauge generators $XIIIIII$, $ZIIIIII$, $IXIIII$, $IZIIII$, $IIXIIII$, $IIZIIII$. Protected logical representatives are $IIYZIZI$ and $IIIXIZIZ$. Pauli phases are quotiented out; 720 counts symplectic maps, not the full Clifford group including Pauli phases.

Verification route. Enumerate a family, test preservation of the two-block gauge subspace, descend accepted maps to the protected logical quotient, and classify their logical action. The supplied identical-map output records only logical identity and block exchange; the pair-specific searches record the counts above.

Evidence status. The original summary and search outputs support this statement. These entangler searches were not rerun in release 0.5. This is a conditional summary of a reported exhaustive computation, not a newly verified universal theorem.

Limit. No conclusion is established here for arbitrary multi-round circuits, non-Clifford gates, ancilla-assisted methods, code switching, or temporary-gauge parity surgery. A complete noisy logical CNOT remains unestablished. The parity-surgery proposal is not invalidated by this restricted obstruction.

Sources. Supplied transversal-entangler lock and two search-output records, identified in `PROVENANCE.json`. Formal background: Poulin, arXiv:quant-ph/0508131; Gottesman, arXiv:quant-ph/9705052.